SmartGridXAI:
Explainable AI for Electricity Demand Prediction

Energy Demand Forecasting using Smart Meter Data

Problem Statement

Forecasting How Much Electricity is Needed (aka electric demand forecast) is a huge challenge these days for energy systems. Energy suppliers need to be able to accurately predict how much electricity will be used so they can:

- Produce enough power
- Efficiently manage grid load
- Cut down on wasted electricity
- Stop power outages

The way most forecasts are done now, stakeholders can't even see how their electricity use affects the "forecast," as they can't identify why there is an increase or decrease in demand. A predictive system that is both accurate and explainable is greatly needed.

Project Objective

The goal of this initiative is to develop an Explainable AI (XAI) system for predicting power usage and explaining how the predictions are made. The goals of this project are:

01

Build a predictive electric load machine learning algorithm

02

Analyze the variables impacting load

03

Use Explainable AI technologies to provide explanations for model predictions

04

Improve decision making related to energy management

This project will result in the delivery of what we will refer to as **SmartGridXAI**.

Dataset Overview

The dataset used for this research is based on records of electric usage (from smart meters) with various temporal & environmental variables recorded at the same time as the electric usage. The key variables in the dataset are:

- Temperature
- Humidity
- Wind Speed
- Average Past Consumption
- Hour of Day
- Day of Week

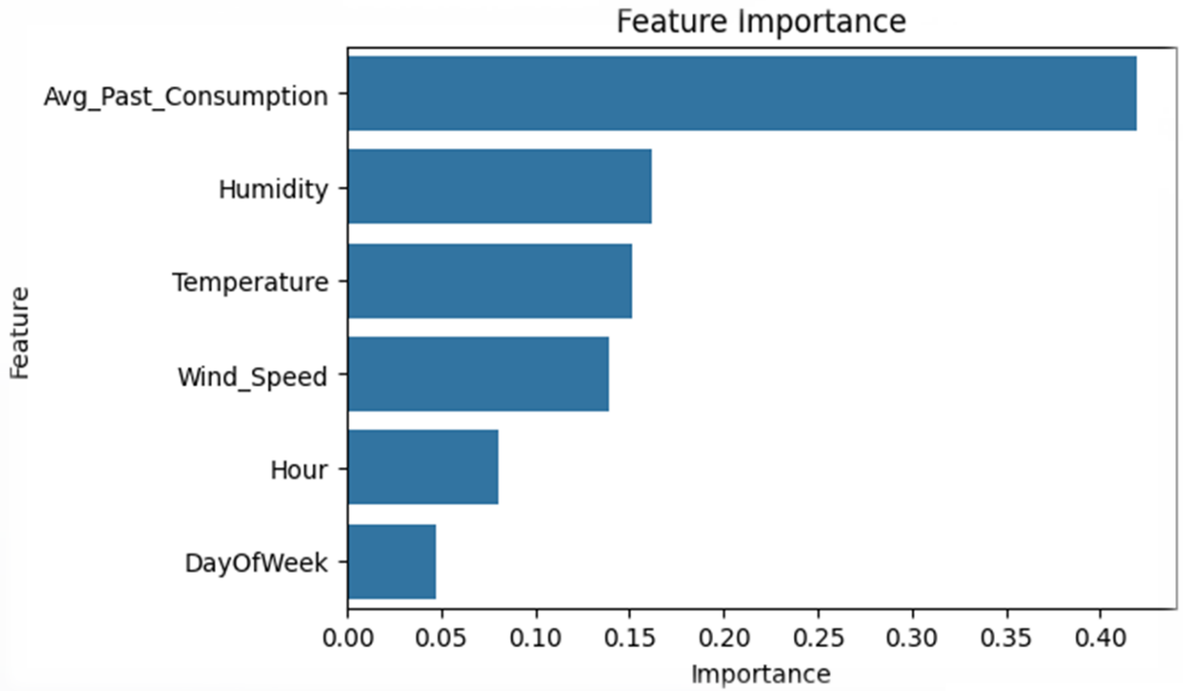
The variable of interest (i.e., the target variable) is **Electricity_Consumed (in kWh)**.

The model uses various combinations of these features to identify environmental conditions (e.g. weather, temperature) and the effect of human behaviour on electric consumption (i.e., the time of day, or the day of the week).

Data Exploration

During this study there were trends that could be identified through Exploratory Data Analysis using this dataset. The key take-aways are:

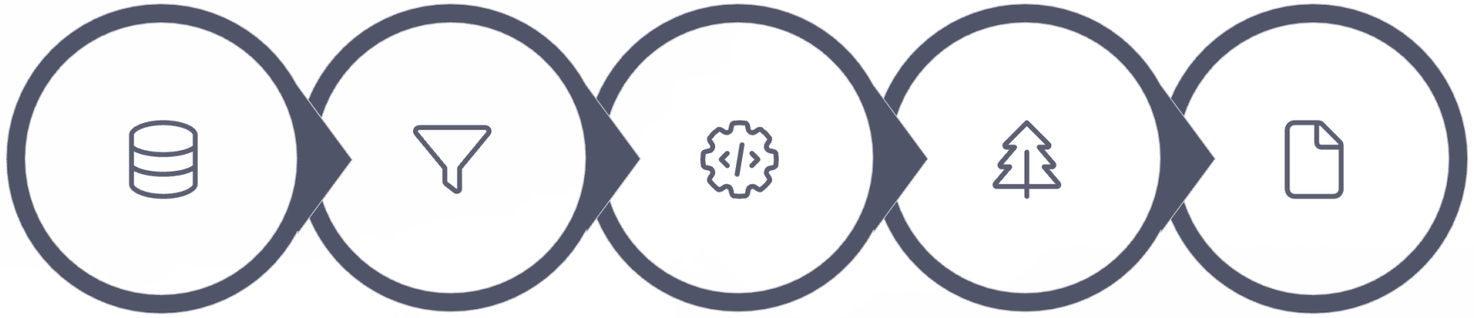
- The amount of power consumed by any location in a given day can vary greatly based on the time of day.
- Power usage trends show that historical usage patterns are correlated to the future rate at which that power will be used.
- Factors that fall under the category of the environment (temperature, humidity, etc.) have an influence on how much power that location is going to use at any time.



These findings were used during the feature engineering and model building phases.

Methodology

This project adheres to a structured workflow of machine learning:

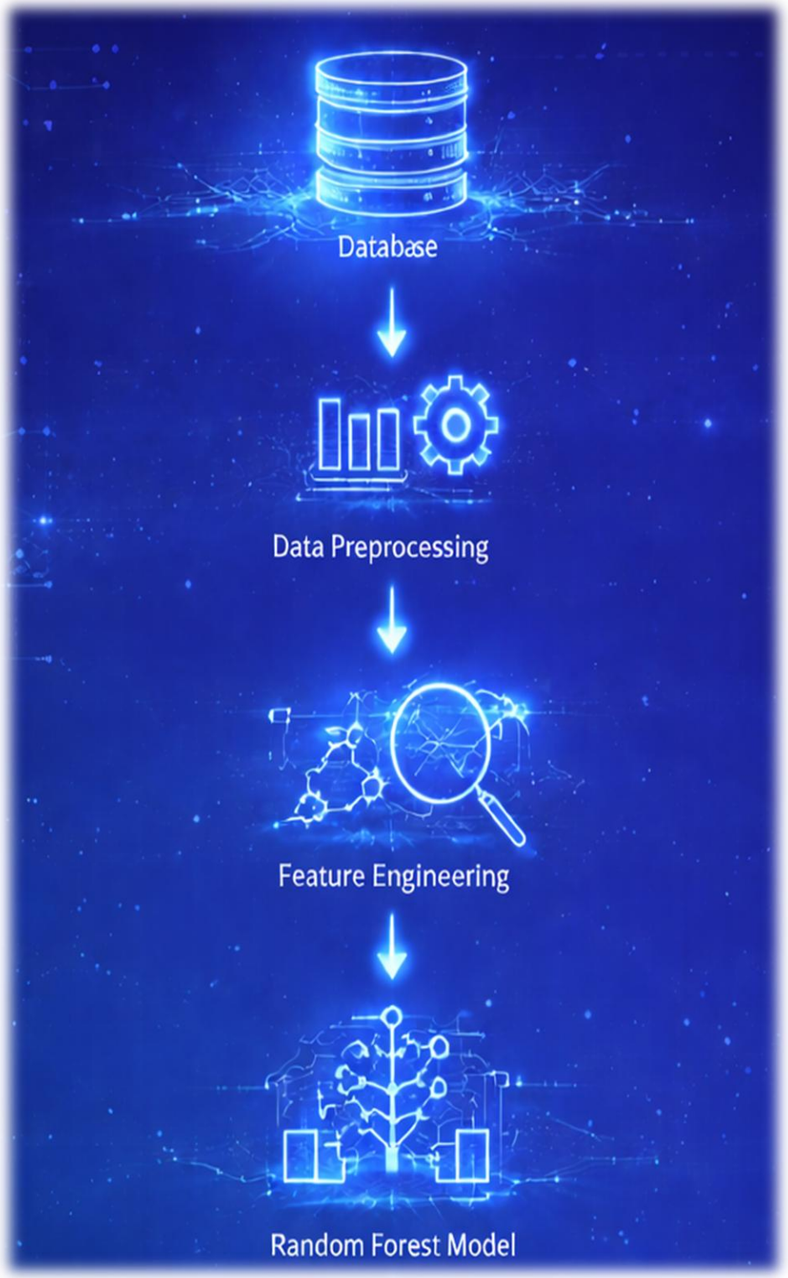


Data set | Data Preprocessing | Feature Engineering | Random Forest | Explainable AI

Worked on with the following technologies:

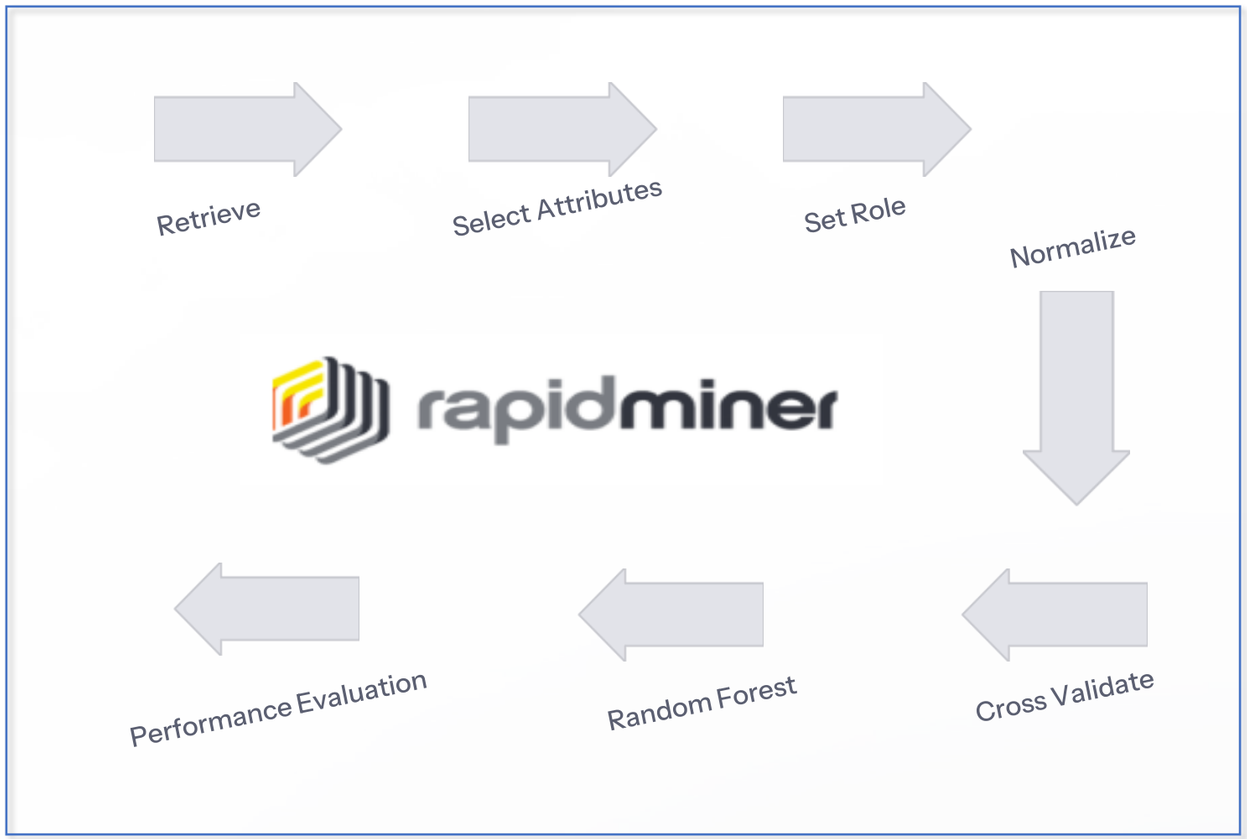
Python	RapidMiner	LIME
Scikit-learn	SHAP	

By using this combination, the project provides not only predictive models but also the ability to interpret them.



RapidMiner Integration

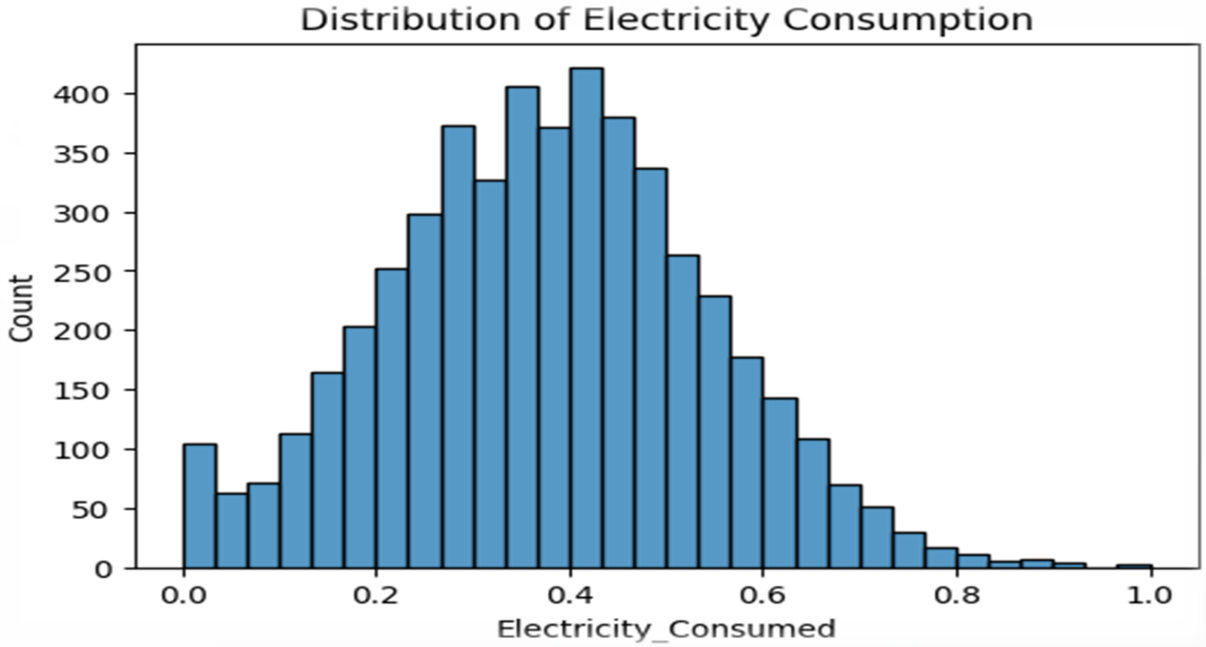
The visual machine learning pipeline created with RapidMiner includes a series of sequential steps for retrieving data, preparing data for analysis, selecting features from the prepared dataset, selecting and normalizing features, evaluating the performance of the Random Forest model, and performing cross-validation of the Random Forest model in order to determine its robustness. The workflow in the pipeline includes:

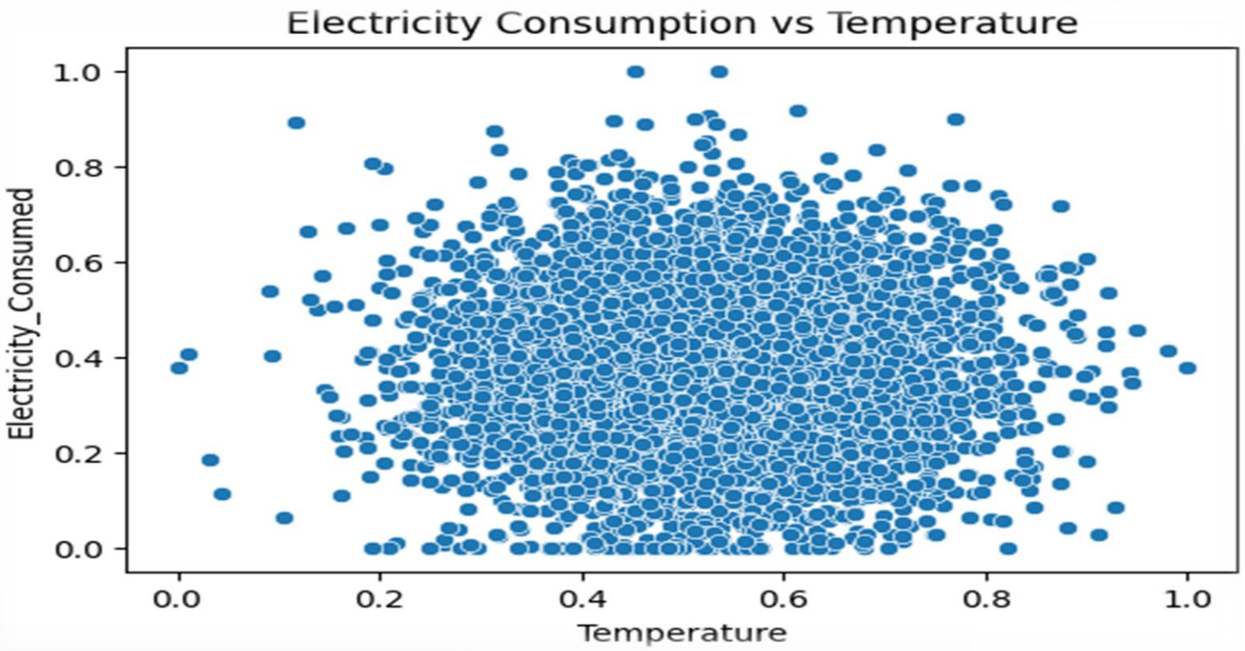


Model Performance

To predict the level of demand for electricity over time, a Random Forest regression model has been developed and evaluated based on three commonly used regression metrics: the Mean Squared Error (MSE), the Mean Absolute Error (MAE), and the R^2 score. These evaluation metrics provide a quantitative measure of how accurate the model is at predicting future electricity demand as well as how well it captures the typical patterns of demand for electricity.

Overall, the random forest regression model provides a successful means of developing the relationships between environmental factors (such as temperature and humidity), past consumption of electricity, and future demand for electricity.





Explainable AI

To enhance model transparency, there were two approaches to developing a model that is interpretable by humans through the use of artificial intelligence.

SHAP

SHapley Additive Explanations

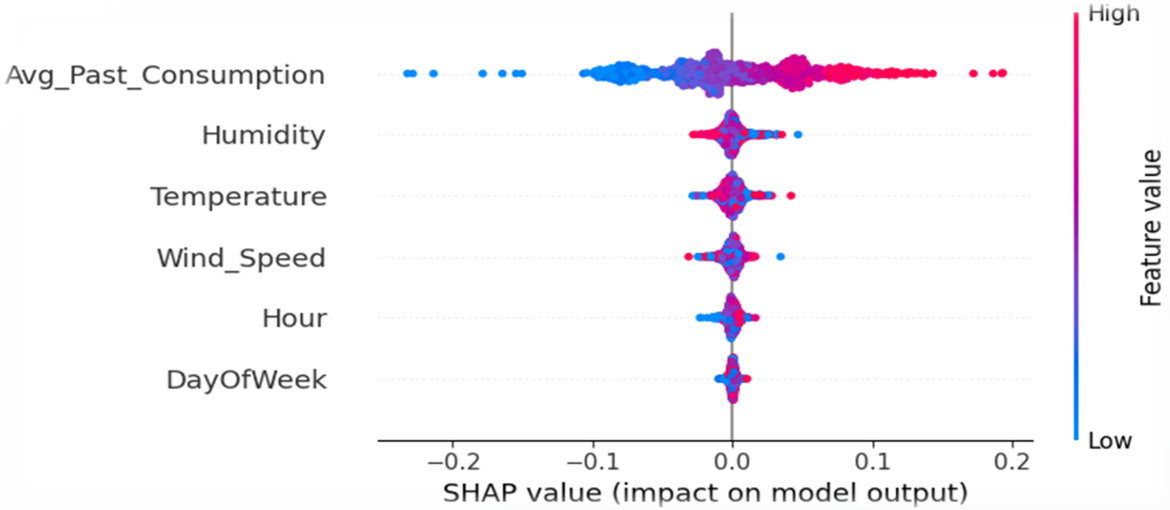
- Provides a ranking of the importance of features globally
- Shows how each feature influences the predicted outcome of each case

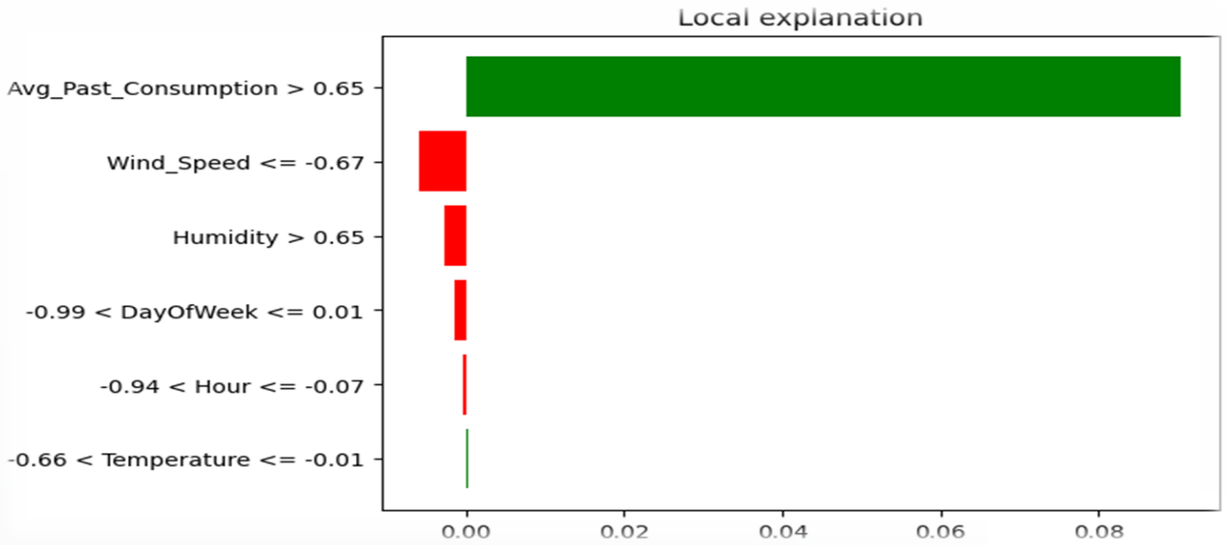
LIME

Local Interpretable Model-Agnostic Explanations

- Explains the various parts influencing each prediction
- Shows the features related to a single prediction

Both of these techniques provide a way for the machine learning model to be interpreted and trusted by decision makers.





Key Insights & Conclusion

The Model has produced many significant insights, including the following:

1

Average Past Usage was the most substantial predictor of future electricity demand.

2

Weather variables, including temperature and humidity, were good indicators of how electricity will be used.

3

Customer electricity usage varies considerably based on time of day – similar to how people behave daily.

4

Energy providers may use these insights to predict future energy consumption patterns and improve grid operation efficiency.

The SmartGridXAI project explores the use of machine learning and explainable AI to enhance predictions related to electricity usage and demand. Its main contributions include:

- an explainable machine learning model for forecasting electricity demand
- integration of an analytics workflow through RapidMiner
- understanding the key factors affecting electricity demand based on data analysis.

One of the main advantages of combining predictive modelling with explainable AI is that it provides a highly scalable and transparent method for conducting smart grid energy analysis. Future enhancements may include integrating this model with real-time energy consumption dashboards and utilizing larger smart grid datasets.